

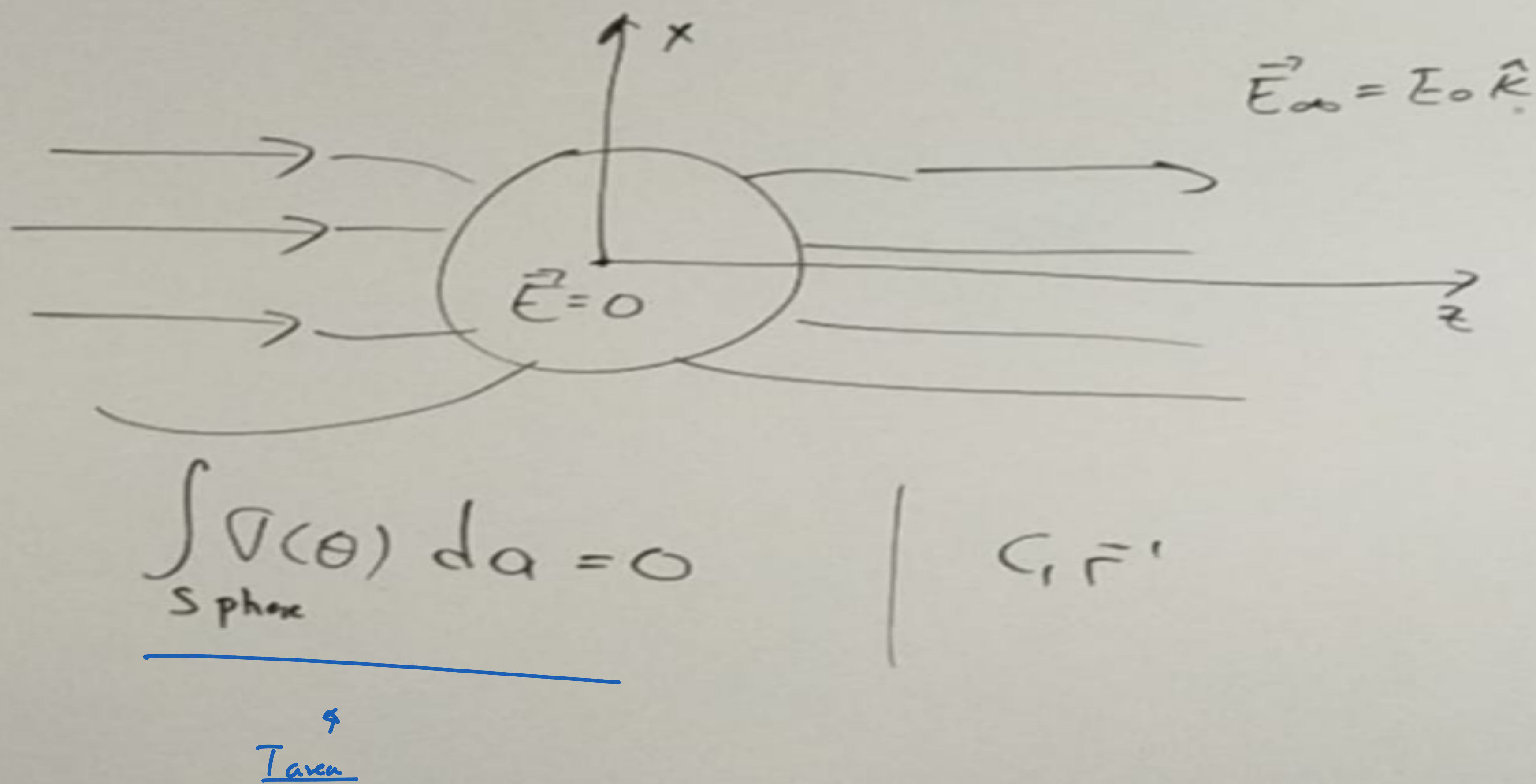
Unit 1: Electrostatics

☑ solutions to Laplace equation

$$\nabla^2 u = 0$$

spherical coordinates (azimuthal symmetry)

$$U = U(r, \theta) = \sum_{k=0}^{\infty} \left[A_k r^k + C_k r^{-(k+1)} \right] P_k(\cos \theta)$$



☑ Laplace equation in cylindrical coordinates

$$U = U(r, \theta)$$

$$\frac{\partial U}{\partial z} = 0$$

$$\nabla^2 U = \frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial U}{\partial r} + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2} + \frac{\partial^2 U}{\partial z^2} = 0$$

$$\Rightarrow \frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial U}{\partial r} + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2} = 0$$

$$U = Y(r)S(\theta) \quad \left. \begin{array}{l} \text{Propose} \\ \text{solutions} \end{array} \right\} \text{separable}$$

Tenemos

$$\frac{1}{r} \frac{\partial}{\partial r} (r) \frac{\partial}{\partial r} (Y(r)S(\theta)) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} (Y(r)S(\theta)) = 0$$

$\Rightarrow$ Tenemos

$$\left[\frac{S}{r} \frac{d}{dr} r \frac{dY}{dr} + \frac{Y}{r^2} \frac{d^2 S}{d\theta^2} = 0 \right] \left(\frac{r^2}{YS} \right)$$

Tenemos

$$\frac{r}{Y} \frac{d}{dr} r \frac{dY}{dr} + \frac{1}{S} \frac{d^2 S}{d\theta^2} = 0$$

$$\frac{S}{r} \frac{d}{dr} r \frac{dY}{dr} + \frac{Y}{r^2} \frac{d^2 S}{d\theta^2} = 0 \quad \left(\frac{r^2}{YS} \right)$$

$$\frac{r}{y} \frac{d}{dr} r \frac{dy}{dr} + \frac{1}{S} \frac{d^2 S}{d\theta^2} = 0$$

K -K

$$\Rightarrow \frac{d^2 S}{d\theta^2} + KS = 0$$

$$\frac{r}{y} \frac{d}{dr} r \frac{dy}{dr} + \frac{1}{S} \frac{d^2 S}{d\theta^2} = 0$$

K -K

$$\Rightarrow \frac{d^2 S}{d\theta^2} + KS = 0$$

$$S(\theta) = \begin{cases} \cos \sqrt{K} \theta \\ \sin \sqrt{K} \theta \end{cases}$$

For radial equation:

$$r \frac{d}{dr} r \frac{dy}{dr} = Ky$$

$$y(r) = r^a \Rightarrow a^2 y = Ky$$

$$y(r) = \begin{cases} r^n \\ r^{-n} \end{cases}$$

Unit 1: Electrostatics

Solutions to Laplace Equation

For radial equation:

$$r \frac{d}{dr} r \frac{dy}{dr} = Ky \quad y(r) = r^a \Rightarrow a^2 y = Ky$$

$$\Rightarrow a^2 = K$$

$$\Rightarrow a = \pm \sqrt{K}$$

$$\Rightarrow y(r) = \begin{cases} r^n \\ r^{-n} \end{cases}$$

$$\text{If } K > 0, \quad K = n^2$$

$$\cos(n\theta)$$

$$\sin(n\theta)$$

$$\cos(n\theta + 2\pi) = \cos(n\theta)$$

$$\left. \begin{aligned} \cos(n\theta) \cos(2\pi) - \sin(n\theta) \sin(2\pi) \\ \cos(n\theta) - \sin(n\theta) = \cos(n\theta) \end{aligned} \right\}$$

$$1 - \sin(n\theta) = 1$$

$$\Rightarrow \sin(n\theta) = 0$$

$$\text{if } K > 0, \quad K = n^2$$

$$\cos(n\theta)$$

$$\sin(n\theta)$$

$$\cos(n(\theta + 2\pi)) = \cos(n\theta)$$

$$\cos(n\theta) \cos(2n\pi) - \sin(n\theta) \sin(2n\pi) = \cos(n\theta)$$

$$n = 1, 2, 3, 4, 5, 6, 7, 8, \dots$$

$$\text{if } K = 0, \text{ then}$$

$$y(r) = \int \ln r$$

$$S(\theta) = \begin{cases} \theta \\ 1 \end{cases}$$

Tenemos

$$r \frac{d}{dr} r \frac{dy}{dr} = 0 \Rightarrow r \frac{dy}{dr} = A \Rightarrow \frac{dy}{dr} = \frac{A}{r}$$

$$\hookrightarrow y = A \ln r + B$$

$$\frac{d^2 s}{d\theta^2} = 0 \Rightarrow \frac{ds}{d\theta} = A \Rightarrow s = A\theta + B$$

Forma general.

*

$$U(r, \theta) = \underline{A \ln r} + \underline{B} + \sum_{n=1}^{\infty} r^n (C_n \cos n\theta + D_n \sin n\theta) +$$

$n=0$

$$+ \sum_{n=1}^{\infty} r^{-n} (E_n^* \cos n\theta + F_n \sin n\theta)$$

Ecuación de Laplace en coordenadas cartesianas.

▮ Solution to Laplace equation in Cartesian coordinates.